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To cite this article: Sarah Halliday, Tiffany Lavis, Peta Callaghan & Anna Chur-Hansen (2026) Generative AI in higher education psychology programs: a scoping review exploring the opportunities for its use in assessment methods, Australian Psychologist, 61:2, 115-126, DOI: [10.1080/00050067.2025.2580236](https://doi.org/10.1080/00050067.2025.2580236)

To link to this article: <https://doi.org/10.1080/00050067.2025.2580236>



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REVIEW ARTICLE



Generative AI in higher education psychology programs: a scoping review exploring the opportunities for its use in assessment methods

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ABSTRACT

Objective: The current literature on Generative Artificial Intelligence (GenAI) in tertiary settings primarily focuses on the risk it poses to academic integrity, and ways to reduce or remove GenAI use in assessments. As psychology graduates enter a workforce where GenAI is present, educators need to prepare students to use GenAI responsibly. This scoping review aims to assess the current state of knowledge in tertiary psychology regarding opportunities for integrating GenAI into assessment methods.

Method: A comprehensive literature search identified four studies for inclusion. These were published in Australia, Canada, Switzerland, and the United States, and included two quantitative case studies, a mixed-method case study, and a pedagogical case study.

Results: Three themes were generated: 1) GenAI can be used as an effective psychology tutor, 2) GenAI can be used for authentic assessment in undergraduate psychology, and 3) Critiquing GenAI as a form of assessment can enhance student learning and AI literacy.

Conclusions: Only four studies were identified, but all indicate that GenAI can be meaningfully incorporated into psychology assessments. However, this is an underdeveloped area and ongoing research with a particular focus on developing evidence-based assessment methods which adapt to the evolving GenAI landscape is needed.

KEY POINTS

What is already known about this topic:

- (1) Generative Artificial Intelligence (GenAI) has disrupted the education sector, with educators re-evaluating their teaching practices and tertiary institutions revising their policies.
- (2) Both educators and students use and appreciate the benefits of GenAI for administrative tasks and reducing cognitive load; however, there are ongoing challenges and concerns about its role in student assessment tasks.
- (3) In Australian psychology programs, where students must meet specific standards under accreditation, educators have implemented bans or adopted alternative assessment methods that minimise or completely remove its capability.

What this topic adds:

- (1) This scoping review is the first to report the findings of a comprehensive search that focus on the opportunities of integrating GenAI into tertiary psychology student assessment methods.
- (2) The findings offer examples and suggestions for using GenAI as a personal tutor, incorporating and critically evaluating GenAI to develop students' artificial intelligence literacy, and illustrating how it can be drawn on to develop authentic assessments.
- (3) We highlight opportunities for future research, given the dearth of work to date, focusing on the need to increase the evidence-base and for designing adaptable assessment methods that evolve with GenAI.

ARTICLE HISTORY

Received 2 June 2025

Accepted 17 October 2025

KEYWORDS

Tertiary psychology; higher education; psychology teaching and learning; Generative Artificial Intelligence; assessment methods

Introduction

An overview of Generative Artificial Intelligence (GenAI) in higher education settings

The popularity and accessibility of Generative Artificial Intelligence (GenAI) models, such as ChatGPT developed by OpenAI, have educators re-evaluating their

teaching practices and universities updating their policies (Cotton et al., 2024). GenAI models extend on artificial intelligence models by being able to generate new and original content (in various forms like text, photos, videos) in response to user prompts (Wei et al., 2024). Although artificial intelligence has been around for nearly 70 years (Crawford et al., 2023), recent

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 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/00050067.2025.2580236>

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advancements in large language models (LLMs) and the availability of big datasets for training have transformed its applications. These LLMs exceed in producing conversational-style outputs, providing a natural, human-like response to a wide range of text-based queries (Wei et al., 2024). Combining GenAI and LLMs ultimately allows for such models to be more accessible, enabling the general public to use and understand the technology more easily. Introduced in 2022, ChatGPT quickly (i.e., gained 100 million users in 2 months) became the most popular GenAI and LLM model due to its specific design and ability to respond to prompts in a human-like conversational style (Crawford et al., 2023; Lee et al., 2024). It has the capacity to perform a wide range of language tasks including question answering, text generation, translation, and summarisation to a sophisticated standard (Brown et al., 2020; Lee et al., 2024). The introduction of this tool has disrupted the education sector (Koop & Gröbinger, 2024), and as a result, has implications for the way in which teachers teach, and students learn. In higher education settings specifically, the use of GenAI by educators and students presents unique benefits and challenges, largely in relation to assessment methods.

The role of GenAI in assessment methods

A key concern for educators regarding the use of GenAI in higher education is its impact on student learning, particularly in the context of student assessments (Crawford et al., 2023). Assessments in higher education settings are essential to measure learning and ensure students are achieving course and program outcomes (Mislevy et al., 2003). Over time assessments have evolved to meet and incorporate technological capabilities; however, the introduction of GenAI has arguably presented the most unique opportunities for improvement, but also some of the most complex issues that need addressing, to date (Kizilcec et al., 2024). With 83% of Australian university students using ChatGPT for their studies (Chung et al., 2024), addressing GenAI in higher education assessment has become a timely, pressing issue.

While GenAI has prompted educators to rethink and change their practices, doing so has encouraged creative and innovative methods of assessment design and delivery (Xia et al., 2024). A concerted effort has been made to shift the focus of assessing the product to assessing the process, and in doing so, providing students the opportunity to reflect on their learning and develop meta-cognitive skills (Mulder et al., 2023). Examples of this in practice include self-reflection

exercises and logbooks or portfolios for assessing learning through the steps taken over time (Mulder et al., 2023). There has also been a growing emphasis on the importance of authentic assessment by encouraging the application of knowledge and challenging students' beliefs and critical thinking skills (Crawford et al., 2023; Xia et al., 2024). While GenAI has provided an opportunity for educators to be creative in assessment design, there remain substantial concerns for well-established assessment methods in higher education settings.

One of the most pressing issues is the potential for GenAI to be used by students to complete summative assessments. Given GenAI models, such as ChatGPT, can answer questions with a nearly human-level performance (Krause & Stolzenburg, 2023), it is becoming difficult to determine if a student has utilised this software to complete an assessment task, which raises academic integrity concerns such as plagiarism (Cotton et al., 2024). For instance, if a student were to use ChatGPT to generate an essay based on the provided prompt and submit it as their own work, doing so would be considered cheating, and thus, an academic integrity breach. It can be argued that using ChatGPT to bypass completing assessment tasks is stifling the development of critical thinking, problem solving, and creativity skills; ultimately undermining the purpose of higher education (Cotton et al., 2024; Dien, 2023; Dwivedi et al., 2023). Additionally, there are plagiarism and ethical concerns in relation to where ChatGPT sources its information, often doing so without appropriate attribution or by fabricating sources entirely (that is, providing inaccurate or non-existent references) (Bhattacharyya et al., 2023; Dien, 2023; Dwivedi et al., 2023). While these inaccuracies are being addressed quickly through regular updates in versions, these version improvements to GenAI and ChatGPT are raising concerns for other methods of assessment.

At the time of writing, in 2025, GenAI now has the ability to produce a standard of work that is considered acceptable, and in the context of Multiple-Choice Question (MCQ) testing, a standard of work that would pass a summative examination (Newton & Xiromeriti, 2024). A recent scoping review exploring the performance of ChatGPT on MCQs from a wide range of disciplines (medicine, law, computer science, economics, and mathematics) found previous free versions of ChatGPT (ChatGPT3 and ChatGPT3.5) did not always perform well enough to pass exams (an average of 53.6% correct answers across 48,596 items across disciplines) and underperformed compared to the average student (Newton &

Xiomeriti, 2024). However, with the introduction of ChatGPT4, the number of correct answers rose to 75.1% when tested on over 19,000 items, achieving a passing mark for 92.9% of the included examinations. While MCQs serve a purpose as an effective form of assessment for large cohorts of students, it is noteworthy that GenAI has been improved to a point where it now achieves a passing grade or better, in exams. The risks GenAI poses for assessment methods are also problematic across disciplines, including undergraduate and postgraduate psychology programs.

GenAI in tertiary psychology

In Australia, the Australian Psychology Accreditation Council (APAC) ensures psychology programs in higher education settings maintain a standard in the education and training of future psychologists (Australian Psychology Accreditation Council, 2019). As part of this process, APAC regularly reviews psychology programs for quality assurance and quality improvement against Accreditation Standards; the guidelines which ensure that graduates of accredited programs are well-equipped to employ their psychological knowledge and skills in the community. While maintaining the APAC standards for psychology graduates is essential, the prevalence of GenAI use among university students raises concerns about whether psychology graduates truly acquire the skills and knowledge being taught and assessed (Kizilcec et al., 2024). This, in turn, raises questions about how such knowledge is applied and ethically disseminated in society (Kizilcec et al., 2024). These concerns are important in the field of psychology, where GenAI can provide appropriate responses specific to the psychology discipline.

Machin et al. (2024) recently explored the ability of GenAI for psychology-specific assessments, explicitly whether the tool could demonstrate psychological literacy; the ethical application of psychological skills and knowledge which APAC requires students to achieve (Australian Psychology Accreditation Council, 2019; Murdoch, 2016). Findings from 13 prompts based on research methods scenarios determined that ChatGPT responded in ways that would typically be regarded as displaying a high level of psychological literacy. The accuracy of this tool in the research methods domain raises questions about its ability to generalise to other domains of psychological literacy, or other knowledge and skills required in an APAC accredited course. The risk of students using GenAI to complete work and therefore not demonstrating the qualities of a psychology graduate has prompted educators to

either ban the use of GenAI in their courses completely or become creative and consider alternative forms of assessment which circumvent GenAI's capability (Cotton et al., 2024).

One such example is drawing on interactive oral presentations as a form of assessment which reduces the potential for academic misconduct, requiring students to engage in a "genuine and unscripted interaction between a student ... and an examiner ... demonstrat[ing] knowledge verbally in an authentic setting, representative of what would be encountered in the workplace" (Sotiriadou et al., 2020, pp. 2132–2133). Newell (2023) explored this method in the context of psychology and provided comprehensive considerations for implementing interactive oral assessments in psychology programs, and made recommendations to address scalability challenges given psychology courses often have a large cohort of students (Quality Indicators for Learning and Teaching [QILT], 2024). Solutions included having multiple assessors for marking, a rotating panel, or utilising technology to facilitate an online interactive oral assessment (Newell, 2023). While it is encouraging to see innovative assessment methods developed to mitigate the risks posed by GenAI, it is equally important to explore ways to integrate or collaborate with these tools in psychology student assessments. As graduates enter workplaces where GenAI is readily available, educational settings should equip students with an understanding of the strengths, limitations, and the biases of these tools (Johnston et al., 2024; Yu, 2023). This exposure will help them navigate and use the technology responsibly in their future careers.

The current review

Research on GenAI in psychology assessments is relatively new. There is a need to bring together current knowledge to provide clear and informed recommendations for future research and teaching practice. Additionally, existing literature primarily focuses on the risk GenAI poses to academic integrity, or how it can be reduced or removed completely by assessing students through other methods. While these concerns remain valid, the presence of GenAI now, and into the future, necessitates a shift of focus. In Australian psychology programs, educators are responsible for ensuring graduates enter the workforce equipped to apply their psychological knowledge and skills. With GenAI now present in many workplaces, educators are also responsible for preparing students to use this technology ethically. To

achieve this, psychology programs should be moving towards ensuring graduates can responsibly employ psychological knowledge and skills while integrating GenAI into their workplaces. One way to address this is to explore how GenAI can be incorporated appropriately in assessment methods, providing students with an opportunity to practice how to use and responsibly integrate GenAI before entering the workforce. As such, the aims of this scoping review are as follows:

- (1) To determine the current state of knowledge regarding how tertiary psychology educators incorporate GenAI in assessment methods.
- (2) To identify areas where research is lacking to guide future research and practice.

Materials and methods

Design

Scoping reviews are an ideal tool for determining the scope of a body of literature on a given topic, identifying underlying key concepts, and the exploratory nature informs areas for subsequent research (Munn et al., 2018). This review adhered to the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to provide a comprehensive data collection procedure (Page et al., 2021).

Search strategy and eligibility criteria

The overall search strategy, including the databases selected, search terms and search criteria, was created with the assistance of an expert liaison librarian at the University of Adelaide. The databases were selected for their focus on psychology and education and included PsycINFO, Scopus, ERIC, and ProQuest Central. Search terms consisted of the key concepts: psychology education, assessment, and artificial intelligence (see Supplementary materials for the full list). Grey literature in the form of published reports from recognised and reputable psychological associations or universities were also included. Database and grey literature search occurred from 20–22 January 2025, with monitoring occurring until 29 September 2025.

To be eligible for inclusion studies and reports needed to focus on how educators in tertiary psychology (undergraduate or postgraduate) used Generative Artificial Intelligence (GenAI) in the form of student assessment. This included either having GenAI embedded into the assessment design or allowing GenAI to be used as a support to improve assessment

performance. As this review specifically examined the use of GenAI in psychology student assessment methods, studies were excluded if their primary focus was on circumventing the risks associated with GenAI by proposing alternative assessment methods (see Newell (2023) as an example). Studies could have a qualitative, quantitative, or mixed-method study design, could be a case study or pedagogical discussion piece, and had to be written in English.

Study selection and synthesis

A total of 98 records were identified in the first database and grey literature search, with 23 of those duplicates. From the 75 records screened by title and abstract, 65 were excluded with the most common exclusion being related to the absence of GenAI in the paper. A subset of 10% of records in this initial screening was reviewed by two members of the research team to ensure consistency and reliability. There were no discrepancies that required further consultation from other research team members. A total of 10 full-text records were assessed for eligibility, with three of those reviewed by all four team members to ensure consistency. A final number of four records were deemed eligible for inclusion. Figure 1 shows the full screening process. As scoping reviews are designed to provide an overview of the existing evidence base regardless of the quality of evidence, a formal quality appraisal was not performed.

To synthesise and describe the eligible records, a narrative approach was taken to ensure findings were detailed in a logical and articulate manner (Popay et al., 2006). This involved reporting on the key findings of each paper, along with documenting the author(s), year, title, study design, sample population, location, learning context, and what GenAI tool was used. Findings of each paper have been grouped and reported on according to common concepts and themes.

Results

A total of four peer-reviewed articles met the criteria for inclusion in the present study. This small number reflects the recency of the introduction of GenAI into educational settings, especially in the discipline of psychology. Table 1 details the key findings of each paper, along with the author(s), year, title, features of the study design and methods, and which GenAI tool was used. The papers were based in Canada (Chambers & Owen, 2024), Australia (Richmond & Nicholls, 2024), Switzerland (Baillifard et al., 2025), and the United

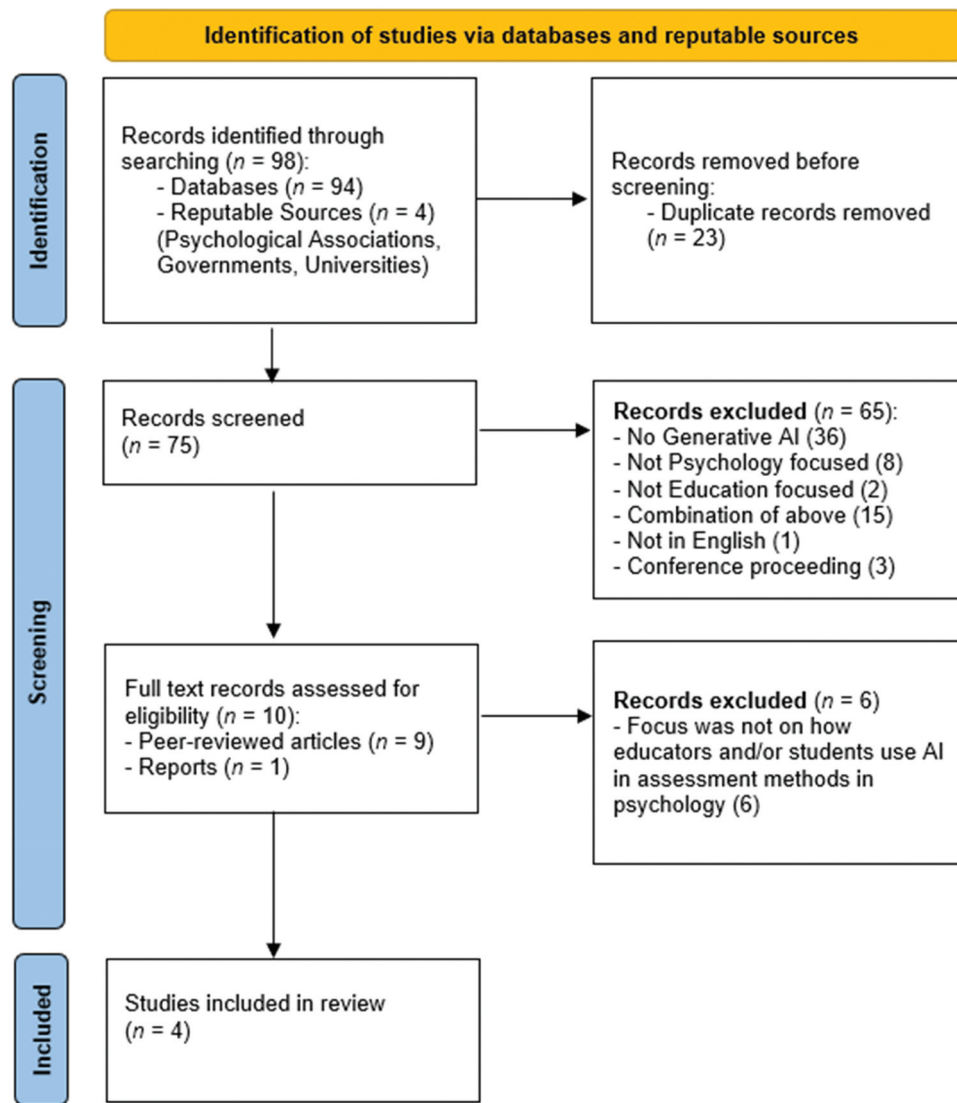


Figure 1. Prisma flowchart detailing findings from the literature search.

States (Miserandino, 2024) and were published either in 2024 or 2025. Common themes identified in these papers are as follows: 1) GenAI can be used as an effective psychology tutor, 2) GenAI can be used for authentic assessment in undergraduate psychology, and 3) Critiquing GenAI as a form of assessment can enhance student learning and literacy.

GenAI can be used as an effective psychology tutor

Two of the included papers examined the effectiveness of using GenAI as a personal study tutor (Baillifard et al., 2025; Chambers & Owen, 2024). As part of an online Neuropsychology and Neurosciences course at UniDistance Suisse, Baillifard et al. (2025) utilised a GenAI tutor developed by MAGMA Learning (<https://www.magmalearning.com>) to provide

personalised assistance in the consolidation of key concepts through retrieval practice and spaced repetition. Throughout the semester, 51 students engaged with the GenAI Tutor at a frequency, duration, and timing that suited them. To determine whether the personalised GenAI Tutor had any impact, final exam results were compared to a similar course running simultaneously that did not offer the GenAI Tutor (Neuroanatomy). Students who actively used the GenAI Tutor for the Neuropsychology and Neurosciences course achieved significantly higher grades on the final exam than those who were inactive. Additionally, active students also improved their ranking by up to 15 percentile points in the Neuropsychology and Neurosciences course relative to their performance in the Neuroanatomy course. While results supported the effectiveness of

Table 1. Summary of eligible studies.

Author(s) (Year)	Title	Study Design	Sample Population, Location and Context (e.g., online or on-campus)	GenAI Tool Used	Key Findings
Baillifard et al. (2025)	Effective learning with a personal AI tutor: A case study	Quantitative Case Study	51 students enrolled in the undergraduate psychology course "Neuropsychology and Neurosciences" UniDistance Suisse, Switzerland Online – Distance Learning format	AI Tutor App developed by MAGMA Learning (https://www.magmalearning.com) Provides a personalised approach to retrieval practice and spaced repetition to assist with consolidating key concepts for the long term.	Students who actively engaged with the AI tutor app achieved significantly higher grades on the final exam compared to non-active students in the Neuropsychology and Neurosciences course. Students who actively engaged with the AI tutor app performed better on the final exam in the course where the app was offered, compared to a similar course where the app was not offered.
Chambers and Owen (2024)	The efficacy of GenAI tools in postsecondary education	Mixed Methods Case Study	40 students enrolled in one of two postsecondary institutions in British Columbia; 14 were native English speakers and the others spoke an Indian language. 18 participants were face to face and 22 attended online course, in introductory psychology	GenAI chatbots (such as ChatGPT) as supplementary study confidantes for exam preparation and as facilitators for an essay grading task.	Objectives were to assess the efficacy of self-testing facilitated by chatbots in enhancing learning processes and, thus, potentially conferring cognitive advantages to students and using written material produced by chatbots to learn about the structure of academic writing. It was also expected that test anxiety would decrease with the use of chatbots. The post-course survey demonstrated positive perceptions of chatbot assistance in understanding course material and constructing academic papers. ChatGPT was the most common tool used but not the only GenAI tool that students chose to use for two assignments. Some other chatbots of choice were Tiimo Vercel, My AI, essaywriters.AI, and Microsoft Bing. The integration of chatbots as "study confidantes" and essay writers was perceived by students to positively influence comprehension and retention of course material. Test anxiety remained relatively stable at pre and post testing, which was not expected – it was expected that anxiety would decrease. The findings underscore the potential of chatbots in improving learners' understanding of writing an academic essay.

(Continued)

Table 1. (Continued).

Author(s) (Year)	Title	Study Design	Sample Population, Location and Context (e.g., online or on-campus)	GenAI Tool Used	Key Findings
Miserandino (2024)	Authentic and creative assessment in a world with AI	Pedagogical discussion/ examples	No exact sample but ideas from personal experience/discussions with colleagues; findings suitable for undergrad psychology programs United States	LLMs: ChatGPT and Latimer, the Black GPT AI image tools: ideogram, Stable Diffusion, Craiyon, Imagine, Leonardo	Offer 5 examples of how to use AI for authentic assessment: (1) Create and correct a misunderstanding: students identify something they don't understand, ask AI to explain it, analyse the AI response, reflect on this process (2) Critical and creative prompting: students generate prompts form unusual perspectives, analyse the responses, compare perspectives, report findings through discussion or written report (3) Critical analysis of AI biases: students use image generators and input prompts to create images based on social categories, compare results across different tools and analyse patterns, class discussions critiquing AIs training data and discuss ways to mitigate bias (4) Understanding systemic biases in AI: students craft AI prompts to test for bias in text generation; analyse how AI-generated responses differ based on subtle prompt variations; discuss findings, speculate on AI-training data, propose solutions to mitigate biases (5) Enhancing employability with mock job application: use AI to create job description for psych-related career; write cover letter, resume, and prepare for mock interview; critique the materials, refine using AI, receive peer or instructor feedback
Richmond and Nicholls (2024)	Using generative AI to promote psychological, feedback, and artificial intelligence literacies in undergraduate psychology	Quantitative	363 undergraduates enrolled in a second-year undergraduate course "Social and Developmental Psychology", Term 1 2023 (February – May) On-Campus – University of New South Wales, Australia	Students were asked to use ChatGPT (or another GenAI tool) to write the first draft of their media release.	Students provided feedback against rubric criteria for the media release written with the assistance of AI. "Style" was identified as a strength of the AI generated output. However, "comprehensively and accurately covering what the goal of the research was, what the researchers did and what they found" was identified as a poor aspect of the AI output. Students were able to accurately identify strengths and weaknesses associated with the AI output, against the rubric criteria. The introduction of the AI critique (2023) improved student performance on the "script revision task" (when compared with 2022 performance) but had no significant impact on performance for the "final video task".

personalised GenAI Tutors in enhancing academic performance in this study, it is important to note that both the GenAI Tutor's learning materials and the final exams were based on rote learning. Therefore, it remains unclear whether personalised GenAI Tutors would be effective for other learning methods used in psychology.

Chambers and Owen (2024) examined the efficacy of chatbots, a type of GenAI tool, to improve learning outcomes in a group of 40 students in an Introduction to Psychology course offered at both the University of Northern British Columbia (UNBC) and the College of New Caledonia (CNC) during the Spring Intersession of 2023. Fourteen of the students spoke English as their first language – the remainder spoke an Indian language as their first language. The chatbots served as “study confidants’ for exam preparation and as facilitators for an essay grading task” (p. 57). The chatbots were used by students as a method of self-testing to prepare for the examination, as well as producing written material for students to learn about the structure of academic writing. The researchers also assessed whether the use of chatbots decreased students’ anxiety in the assessment tasks. The results of the study, assessed through self-report surveys and an anxiety scale, showed that most, but not all students, found the chatbots helpful in preparing for assessments, although this was more pronounced for students whose first language was not English. Pre- and post-assessment anxiety scores were not affected by using chatbots. Whilst interesting, the sample size was small and there were methodological issues which weaken conclusions, such as the reliance on student self-report. It is also unclear from the methodology whether the researchers were the students’ teachers and assessors and whether the data were collected blindly, which poses ethical questions and confounding of roles.

GenAI can be used for authentic assessment in undergraduate psychology

Miserandino (2024) explored how educators can incorporate GenAI into authentic assessments that align with the American Psychological Association’s (APA) learning goals for undergraduate psychology: content knowledge, scientific inquiry, ethical values, communication and technological skills, and professional development (American Psychological Association, 2023). Miserandino (2024) suggests that educators can respond to these goals by designing authentic assessments that engage students in real-

world tasks using GenAI. Activities include using GenAI to clarify misconceptions, critique bias in GenAI outputs, simulate job applications, and generate creative prompts. These tasks can support critical thinking, ethical awareness, collaboration, and reflective learning, positioning GenAI not as a short-cut, but as a tool to enhance student engagement and support skill development aligned with disciplinary standards. Importantly, the article is not empirical; it is a conceptual and pedagogical essay drawing from the existing literature, practical teaching experience, and alignment with APA curricular goals. However, the paper’s strength lies in its practical orientation and alignment with current concerns about GenAI in education. The author acknowledges a lack of empirical evidence on the effectiveness of these approaches, calling for future research to evaluate their pedagogical impact.

Critiquing GenAI as a form of assessment can enhance student learning and literacy

Richmond and Nicholls (2024) explored how GenAI (such as ChatGPT) might be used as part of a psychological literacy assessment task. A total of 363 students participated in the study as part of their second-year psychology course at the University of New South Wales, Australia (UNSW) during Term 1 2023 (February–May). Students used GenAI to create a media release, which was then assessed according to rubric criteria, to determine whether the students were able to identify aspects of the output which might be identified as strengths and weaknesses. Students were encouraged to use the GenAI media release to inform subsequent script revision and video tasks. The results of the study indicate that students identified “style” as a strength of GenAI generated output. However, “comprehensively and accurately covering what the goal of the research was, what the researchers did and what they found” (p. 4) was identified to have been done poorly by GenAI. A comparison (i.e., through grades allocated by instructors) with previous student cohorts who did not have GenAI integrated into the task indicated that the use of GenAI was beneficial, although there was no significant impact to performance on the subsequent video task. The study did not provide any detail regarding whether there were any cohort differences, beyond awareness by the markers that students had used GenAI as part of their assessment preparation.

Discussion

Overall, this scoping review demonstrates that there is much work needed to enhance our understanding of how GenAI can be used in tertiary psychology assessment methods. This is the first review to highlight the potential opportunities of using GenAI in psychology student assessment methods, rather than focusing on risks or ways to mitigate the influence of GenAI. From a systematic search of the literature, a total of only four journal articles were identified and included, which detail how psychology educators and students can draw on, and use, GenAI for various methods of assessment. The included papers outline how GenAI can be used as an effective psychology tutor, how to draw on GenAI to develop authentic assessments, and how to design assessments which include critiquing GenAI, to improve student literacy and learning.

Two studies identified the potential effectiveness of using GenAI as a personalised psychology tutor (Baillifard et al., 2025; Chambers & Owen, 2024). Both studies found that psychology students who used GenAI as a form of personalised tutor felt more prepared for assessments (Chambers & Owen, 2024) and performed better on final exams (Baillifard et al., 2025). These findings mirror those from other disciplines, and recognise the benefits of immediate and tailored feedback, improved academic performance, and the lightened administrative load for educators (Al-Abri, 2025; Celik et al., 2022; C. Chan & Hu, 2023). While the findings from Baillifard et al. (2025) supported the effectiveness of personalised GenAI tutors in enhancing academic performance, it is important to note that the sample size was small, and both the GenAI tutor's learning materials and the final exams were based on rote learning. Rote learning is a memorisation technique that relies on repetition rather than fostering a deeper understanding on a topic (Stevens & Bernier, 2013). This approach tends to result in poorer retention, limited comprehension on a topic, and a lack of critical thinking development (Roche et al., 2009). Additionally, Chambers and Owen (2024) found that personalised GenAI tutors may not have an effect on assessment anxiety; however, given the small sample size, it is worth exploring this relationship further.

Richmond and Nicholls (2024) found undergraduate psychology students were able to recognise the pitfalls and subsequently critique the outputs from GenAI as a form of assessment. Students could identify the weaknesses of GenAI, including the inaccuracies and lack of in-depth understanding of psychological

research, ultimately improving student artificial intelligence literacy. As GenAI continues to be used in the workplace, it is important for tertiary institutions to equip students with the knowledge and skills of GenAI's strengths and limitations (Johnston et al., 2024; Yu, 2023). Undergraduate students acknowledge that GenAI will likely feature in their future careers (Johnston et al., 2024; Knobloch et al., 2024). To prepare students for the workplace, studies such as this one provide an insightful consideration into how GenAI can be incorporated into psychology assessment methods. This research also identified the potential limitations of the usefulness of GenAI, noting that the critique improved student performance for one assessment task (i.e., script revision) and did not significantly impact another assessment task (i.e., video performance). This highlights the need for further research which considers the nuanced assessments used within psychology.

It is well established that authentic assessments are considered a best practice assessment method in undergraduate psychology, largely due to their aim of promoting abilities related to future careers and assessing students using performance standards similar to those found in the workplace (Dunn et al., 2020; Halliday et al., 2024; Ishak & Salter, 2017). Miserandino (2024) draws on this knowledge and provides resources from both their own experience and the broader pedagogical literature, to demonstrate how GenAI could be integrated into authentic assessment. Aligning with the APA's learning goals for undergraduate psychology, this paper outlines how to utilise GenAI as a tool for enhancing student engagement and learning. Similar to Richmond and Nicholls (2024), this paper outlines the importance of demonstrating GenAI's pitfalls and biases, and provides examples on how this could be applied in assessments. While this article was not empirical, there is scope to draw on the ideas presented to design authentic assessments and conduct empirical research to evaluate their utility in tertiary psychology settings.

Limitations and directions for future research

This review is the first to examine the opportunistic use of GenAI in tertiary psychology. The novelty of the topic resulted in only four papers meeting the inclusion criteria. While the goal of the current paper was to shift focus from risks to opportunities of GenAI use in psychology assessments, as risks had been comprehensively addressed for tertiary education (please see Chiu et al. 2023) for a detailed review), the limited

number of findings indicates that this field is presently lacking a strong knowledge base. Given the consensus suggests that “AI is here to stay” (European University Association, 2024), we encourage educators to produce research and share findings which support the idea of using GenAI in assessments in collaborative, innovative, evidence-based ways.

Due to the limited number of studies, the discussions are constrained, and recommendations may not be applicable to all tertiary settings. Most notably, the included papers did not address the challenges of aligning GenAI use in student assessments with diverse and evolving university policies. This is an important limitation considering the response to GenAI has varied in university settings across the world. For example, there are some UK universities where the use of GenAI for assignments is classified as academic misconduct, some Australian universities have reverted to paper-based exams to eliminate GenAI use entirely, and others have revised plagiarism policies as GenAI raises academic integrity concerns (C. K. Y. Chan, 2023). These varying policies and practices make it difficult to generalise the findings across tertiary settings in Australia and globally. While investigating GenAI use in psychology programs and the implications on academic integrity was outside the scope of the current review, it would be worthwhile for future research to consider and discuss opportunities and limitations of institutional policies to assist educators in navigating the boundaries of GenAI use in their setting. This could also lead to a more consistent and evidence-based approach to incorporating GenAI in tertiary psychology student assessment methods.

It is also noteworthy that we were unable to identify any papers for inclusion in our review pertaining to postgraduate psychology student assessments; neither postgraduate professional training leading to registration as a psychologist, or PhD assessments. As GenAI becomes more integrated into workplaces, it is important to ensure graduates entering the workforce as researchers and psychologists are equipped with the ability to ethically use GenAI in practice. For example, inherent biases exist in GenAI software and can pose a significant ethical challenge related to reinforcing stereotypes and potentially causing unfair treatment (Salah et al., 2024). In psychology, where acknowledging bias and reflexivity are expected APAC graduate competencies (Australian Psychology Accreditation Council, 2019), it is imperative that students are aware of these limitations. Incorporating critiquing GenAI outputs into assessment tasks can encourage students to critically consider and examine such flaws, thereby practicing reflexivity skills in preparation for appropriate use in the workplace.

GenAI is progressing at an extraordinary rate, meaning the suggestions drawn from this review should be considered in the context of current GenAI capabilities. While specific examples may become outdated, the concepts will likely remain relevant. However, as the technology advances, adaptations may be required, and findings may differ. Future research would benefit from the development of an adaptable psychology assessment framework that incorporates GenAI but is rooted in pedagogical principle rather than relying on specific tools or examples that may become dated. For example, educators drawing on pedagogical frameworks such as TPACK (which refers to the interplay of technical knowledge, pedagogical knowledge, and content knowledge) (Mishra & Koehler, 2006) may help promote the integration and understanding of technological tools (such as GenAI) in learning environments (Kohnke & Zou, 2025).

Conclusion

This review is the first to explore how GenAI is being used in tertiary psychology settings, specifically focusing on the opportunities for its use in student assessment methods. Although there were a small number of included studies, the findings suggest that GenAI can be incorporated into psychology assessments in a meaningful way. Specifically, GenAI can be used as an effective psychology tutor, can be drawn on in assessments to improve student GenAI literacy, and can be integrated in a way that ensures authenticity in assessments. Although generalisations are challenging, due to the small amount of included studies and varied approaches to artificial intelligence policies in institutions, there is clear direction for future research. Research should aim to develop flexible, pedagogically sound assessment methods in psychology, that can adapt to the everchanging GenAI landscape. This will equip educators with the knowledge and tools needed to create contemporary assessments that are effective, evidence-based, and help prepare psychology students and graduates for the future.

Acknowledgements

Thank you to Vikki Langton, who assisted with the searching of the literature and provided guidance on appropriate search terms to use and databases to review.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

Data sharing is not applicable to this article as no new data were created or analysed in this article.

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